



Failure Analysis Report



UvA Master Programme Conservation and Restoration of Cultural Heritage

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1. Introduction

Analysing material failure is crucial to understanding how objects deteriorate and locating possible damage sources, identifying potential causes of damage. This report focuses on a painted metal chandelier object that was damaged significantly over time. Cracking, chipping, rust bubble development, and peeling of painted layers are among the problems that have been noticed. These failures imply that the degradation process has been impacted by both mechanical and chemical causes.

The object description will provide details on the object's structure, materials, and observed failures. The failure analysis will categorize the types of failures and examine the material characteristics. To support this analysis, visual evidence will be evaluated, and failure mechanisms will be linked to mechanical properties. Lastly, conservation approaches will be considered to propose methods for preserving the object and preventing further deterioration.

This study aims to provide a comprehensive understanding of the failure mechanisms affecting the object and offer insights into its conservation and preservation.

2. Object Description

2.1 Object Description



Fig. 1. An overview of painted metal chandelier

The object is a painted metal chandelier, approximately 25 x 40 cm in size, featuring a unique design of a bouquet of wildflowers (Fig. 1). Its provenance is unknown, as it was found discarded on the street around the year 2023. Based on online research, this style is regarded as characteristic of mass-produced chandeliers popular in Europe during the 1960s.¹

Constructed primarily from metal, the chandelier serves as a decorative indoor lighting fixture, likely

¹ "Franse metalen hanglamp bloemen jaren 60," Vount, accessed March 19, 2025, <https://www.vount.nl/product/franse-metalen-hanglamp-bloemen-jaren-60/>.

intended for residential use. There is no documentation or label found, and the material type of metal and paint are unspecified. Its structure includes a central vertical column with an electrical wire running through it, supporting multiple stems that extend outward in various directions. The stems are attached with intricately shaped metal leaves and flower petals, possibly with soldering, painted in vibrant colours including shades of green, blue, yellow, orange, and white. The petals and leaves are individually bent and curved, creating a dynamic, three-dimensional effect that mimics a natural floral arrangement plucked from a field. The chandelier is equipped with four sockets, each designed to hold light bulbs, detailing for an authentic candle appearance.

2.2 Damage Description

The chandelier is covered in an unknown grey dust, with significant paint peeling around the edges. The damage is primarily observed on the surface of the paint; numerous cracks in the paint layer are visible to the unaided eye. While no metal parts are broken or torn, severe corrosion occurs where the paint layer has peeled off, resulting in red powdery grains on the surface. Since it was found on the street, there is no information about what this object has endured or what kind of impact it may have experienced. Therefore, it is possible that overall deformation or bending occurred after its original geometry was manufactured. A detailed examination of the mechanical surface damages: cracking, peeling, and rust bubbles will be explored in the Types of Mechanical Failures and Theories on Failure Modes sections below.

2.3 Material Characteristics

The manufacturer and specifications of this object are unknown, as it lacks labeling. The metal components appear to contain steel based on their magnetic properties and characteristic reddish rust formation (Fig. 2). For this analysis, we will proceed with the assumption that the metal is steel. Similarly, the paint's exact composition remains unidentified (Fig. 3). Based on typical requirements for metal painting, particularly the need for quick drying times to minimize solvent contact with metal, we assume an acrylic paint was used. This aligns with the object's mass-produced nature, as acrylic paint offers cost-effectiveness. However, we cannot exclude the possibility of primers or additional coating layers.

Without chemical analysis to resolve these material uncertainties, this report will focus on mechanical failures resulting from forces rather than deterioration processes driven by the inherent properties of the materials.

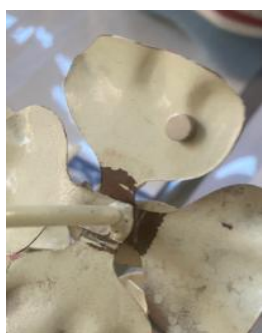


Fig. 2. A magnet sticks to the petal part



Fig. 3. Painted surface

3. Types of Mechanical Failures

3.1 Cracking and Peeling of the Paint Layer

The metal chandelier component exhibits cracks and peeling, mostly along its curved section. In some areas, cracks and peeling occur together; all cracks are accompanied by peeling, visible to the unaided eye, while some chipped regions exist without cracks (Fig.4 and 5).



Fig. 4. Cracks and peelings



Fig. 5. Chipping

3.2 Blistering (Rust Bubbles)

Some areas show blistering. Blistering of paint films on metal manifests as localized delamination, resulting in bubble-like protrusions on the painted surface.² The cause of blistering may vary, but this phenomenon inevitably involves adhesive failure between the paint film and the metal substrate. On this object, blistering areas are accompanied by corrosion of the metal surface (Fig.6).



Fig. 6. Blistering on the surface

4. Theories on Failure Modes

4.1 Stress

4.1.1 Tensile Stress and Compressive Stress on Curvature

When considering only the metal components, the curved floral and leaf-shaped structure induces bending stress during manufacturing. This bending stress leaves tensile, compressive, and residual stresses within the material.

Tensile stress occurs when the substrate's surface stretches in the direction of elongation (Fig. 7). When one side of sheet metal curves outward and elongates, the opposite side contracts. Consequently, the concave inner side experiences a compressive force (Fig. 8). Residual stress is

² Clive H. Hare, "Blistering of Paint Films on Metal, Part I: Osmotic Blistering," *Journal of Protective Coatings and Linings* 15 (1998): 45.

internal stress that remains in the material after manufacturing, processing, or installation.³ During fabrication, as the outer surface undergoes tensile deformation and the inner surface experiences compression, some stress remains internally. Over time, the material attempts to release this stress, potentially leading to material fatigue.

In typical manufacturing processes, paint is applied after the metal is formed. The effects of residual stress released by the metal over time can be observed through gradual deformation of the paint, including adhesion loss. The forces acting on the petal root areas in this object can be attributed to these three stress types. According to Newton's Second Law, force equals mass multiplied by acceleration. For stationary objects with zero acceleration, the net force must also be zero. While the total applied force is zero, individual internal stresses may still exist.

The visible damage, including cracks and peeling of the paint surface, suggests ongoing mechanical stress on the object. Excluding deliberate external forces like impact, these can be explained by residual stress remaining after bending that has not fully released from the metal. Therefore, tensile stress (acting in the elongation direction as metal bends outward) and compressive stress (acting inward on the concave side) likely contribute to the observed damage.

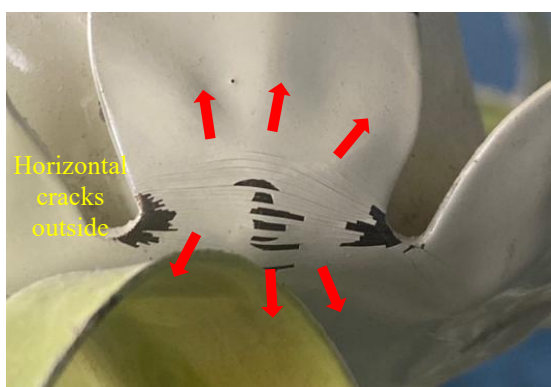


Fig. 7. Outside of curvature
tensile stress applied as red marks

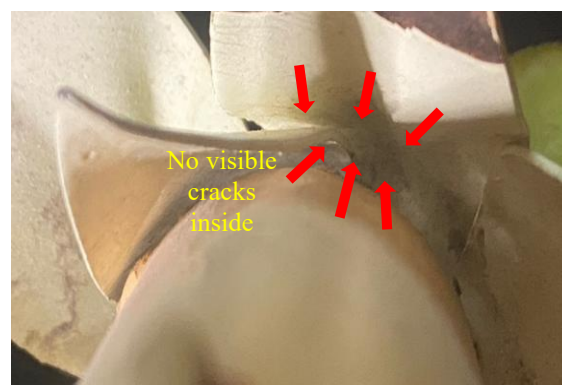


Fig. 8. Inside of curvature
compressive stress applied as red marks
(the same petal with Fig.7)

There is a horizontal crack at the root of the petal (Fig. 7), and although the horizontal line is irregularly bent, the bending direction can be largely divided into upward and downward (Fig. 9). Due to variations in the petal's bending angles, forces are distributed unevenly, resulting in an irregular crack pattern rather than a straight line.

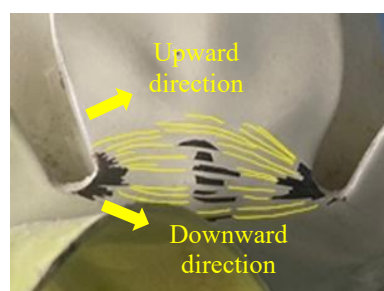


Fig. 9. Irregular crack lines
on the unevenly curved surface

4.1.2 Thermal Expansion

The difference in the coefficient of thermal expansion (CTE) between the paint film and the metal surface plays a crucial role in adhesion. As the temperature fluctuates, these materials expand and

³ "Residual Stress," IMET, accessed March 19, 2025, <https://www.imetllc.com/residual-stress/>.

contract at different rates, weakening the bond between the paint and the metal.

Since the CTE of paint is generally higher than that of metal, temperature variations cause uneven deformation, leading to wrinkles, cracks, and reduced adhesion.⁴ The combination of multi-directional stress and thermal expansion further accelerates adhesion loss. This loss of adhesion can lead to the detachment of the paint layer from the body, especially if cracks isolate areas with weaker adhesion. Ultimately, the paint layer may break into detached fragments.

In other words, the greater the CTE difference between the paint and the metal, the weaker the adhesion. The CTE of acrylic emulsion paint, for instance, ranges from $46\text{--}67 \times 10^{-6} \text{ m/m}^\circ\text{C}$, while that of steel is $10.8\text{--}12.5 \times 10^{-6} \text{ m/m}^\circ\text{C}$.^{5 6} When the temperature rises, the paint expands more than the metal, causing the paint layer to lift or crack. Conversely, when the temperature drops, the paint contracts more than the metal, creating tensile stress that can result in cracking. The important point is that thermal expansion occurs cyclically and repeatedly, eventually causing fatigue in the material.

4.1.3 Corrosion and Volume Change

In areas with blistering, the paint film bulges outward (Fig. 6). This deformation likely occurs because as the metal corrodes, the resulting products expand beyond the original metal volume, pushing against the adhered paint film.

Corrosion initiates when water and oxygen reach the metal surface through micro defects or cracks in the paint caused by various forces. Soluble corrosion products like $\text{Fe}(\text{OH})_2$, ferrous chloride hydrate, and ferrous hydroxy sulphate form beneath the film, generating osmotic pressure.⁷ Once a blister forms, its isolated internal environment accelerates corrosion. The additional volume of rust ultimately pushes the paint layers outwards and causes delamination of the paint film, which was originally applied flush to the metal surface (Fig.10).



Fig. 10. Schematic representation of how the new volume of corrosion pushes the paint layer outwards.

⁴ SinoGraceChem, "Cohesion and Thermal Expansion Coefficient of Paint Film," accessed March 19, 2025, https://www.sinogracechem.com/blog/cohesion-and-thermal-expansion-coefficient-of-paint-film_b162.

⁵ GovInfo, Cohesion and Thermal Expansion Coefficient of Paint Film, accessed March 19, 2025, <https://www.govinfo.gov/content/pkg/GOVPUB-A13-PURL-gpo44537/pdf/GOVPUB-A13-PURL-gpo44537.pdf>, 14.

⁶ Engineering Toolbox, "Linear Expansion Coefficients," accessed March 19, 2025, https://www.engineeringtoolbox.com/linear-expansion-coefficients-d_95.html.

⁷ Hare, "Blistering of Paint Films," 53.

As shown in Fig. 11, the expanding corrosion products push outward against the paint layer where adhesion is weakened, stretching the paint film. If the dried paint is sufficiently ductile, blisters keep their bulging shape; if not, the layer tears, exposing the damage. These tears create additional pathways for corrosive elements to contact the metal, allowing corrosion to continue spreading between the metal surface and the paint layer.

In contrast, relatively fresh chipping areas that display a silvery metal surface without reddish corrosion show no blistering yet (Fig. 12). Disregarding potential chemical bonding-related reactions within the paint itself, the delamination appears to be primarily caused by the mechanical force of expanding corrosion products pushing outward against the paint layer, which weakens adhesion between the metal and the paint layer over time.



Fig. 11. Torn blisters



Fig. 12. Fresh chipping without blistering

4.1.4 Stress Concentration

Notch on Fillet

Cracks initiate from points of stress concentration. When a crack forms in the paint layer and experiences continuous or larger stress, it can lead to the peeling of the paint film. If peeling occurs, it indicates that this area has undergone more severe stress than the cracked area. Examining the peeled paint reveals that the highest stress is concentrated at the notch on the inside corner of the petal (Fig. 13). In certain geometric features, like the notch, they inherently experience more stress. Bill Wei's research shows that the stress in a metal sheet with a notch can be up to four times higher than in a region without a notch (Fig. 14).⁸ While multiple types of tensile stress may act simultaneously on the petal, it is evident that indented areas, such as the peeling paint region, are subjected to higher stress.

⁸ Wei, W. (Bill), and Taylor & Francis, *Art Conservation: Mechanical Properties and Testing of Materials* (Singapore: Jenny Stanford Publishing, 2021), 38.



Fig. 13. Notched area on the petal

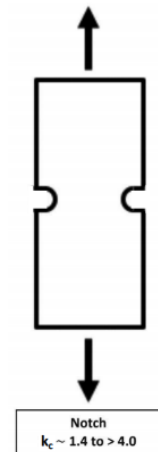


Fig. 14. Diagram for stress concentration of notched area

Radius of the Curvature

The more severe the deformation of the shape, the greater the residual stress will be posed within the material. When comparing two leaves with different angles of curvature, as shown in Fig. 15 and 16, the metal in Fig. 15 is steeply bent, so the residual stress is more significant than in the gently bent part shown in Fig. 16. As a result, cracks and peeling damage occurred inside the concave curvature due to compressive stress. In Fig. 15, the stress concentration on the curvature area was high enough to crack the paint layer, while that in Fig. 16 was not able to make a visible crack.

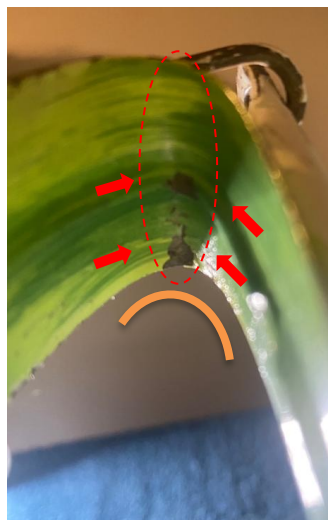


Fig. 15. Smaller radius of curvature with damage



Fig. 16. Larger radius of curvature without damage

In theory, the formula for calculating the stress concentration factor in an object according to its curvature suggests that the smaller the radius of the crack tip (P_t), the higher the stress concentration factor (K_t) (Fig. 17 and 18).⁹ This explanation is related to the crack tip, but the logic that higher

⁹ William D. Allister and David G. Rethwisch, *Materials Science and Engineering: An Introduction*, 9th ed. (Hoboken, NJ: Wiley, 2014), 259.

curvature of the curve results in higher stress can be applied to the degree of curvature of the leaves.

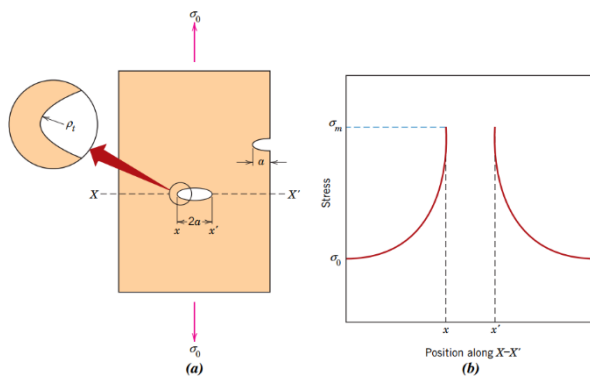


Fig. 17. (a) Demonstrating stress amplification at crack tip positions.
(b) Schematic stress profile along the line X-X' in (a)

$$K_t = \frac{\sigma_m}{\sigma_0} = 2 \left(\frac{a}{\rho_t} \right)^{1/2}$$

Fig. 18. Formula for the stress concentration factor K_t

4.2 Modulus of Elasticity of Aged Paint

It is well known that paint loses moisture as it dries over time. The paint film of the object cracked and eventually peeled off, indicating a loss of ductility needed to endure elongation by stress (Fig.19). For instance, acrylic paint shows a Modulus of Elasticity with respect to aging, approximately as shown below (Fig. 20).



Fig. 19. Brittle fragments of paint film

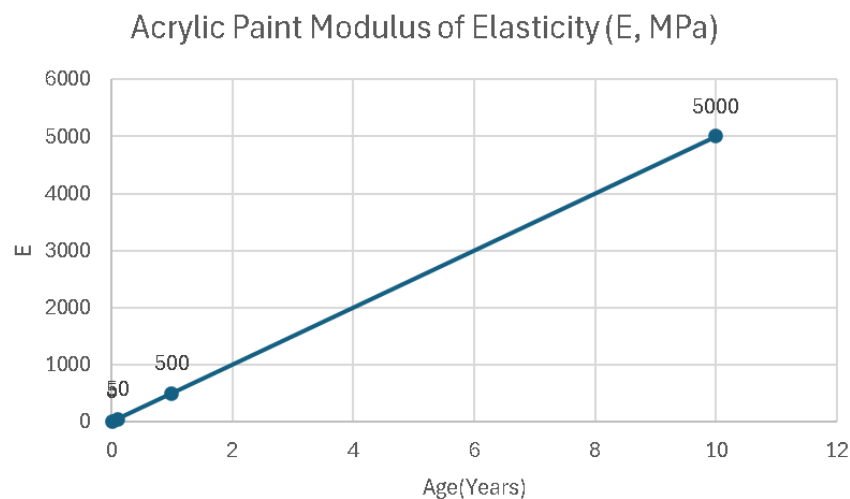


Fig. 20. Graph showing modulus of elasticity (MPa) of Acrylic paint vs aging time. The Graph does not account for testing variables such as, T, RH, Particle types, etc, contributing to the wide spread of the modulus data.¹⁰

The line graph shows a roughly straight upward trend, indicating a steady increase in the modulus of elasticity of acrylic paint over time (Fig. 20). This suggests that acrylic paint becomes more brittle as it ages. Initially, the elastic modulus is low, but over time, it increases to 5000 MPa, making the paint film significantly brittle. Increased brittleness heightens vulnerability to stress, meaning that the paint layer of this chandelier, which is over 50 years old, can easily crack or break more than the fresh paint when the same stress is posed. Based on the graph in Fig. 20, When drawing the stress-strain curves of one-year-old and ten-years-old acrylic paint, it shows that the ten-years-old acrylic paint undergoes only minimal strain before failing and fails at a lower stress level (Fig. 21 and 22).

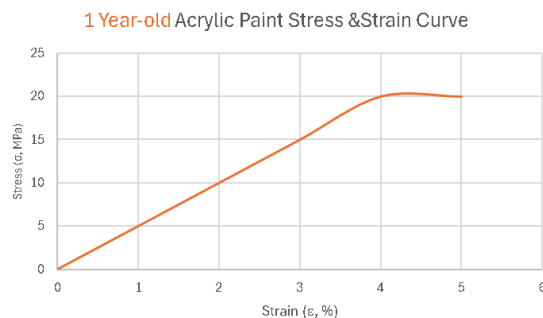


Fig. 21. Hypothetical one-year-old acrylic paint stress-strain curve

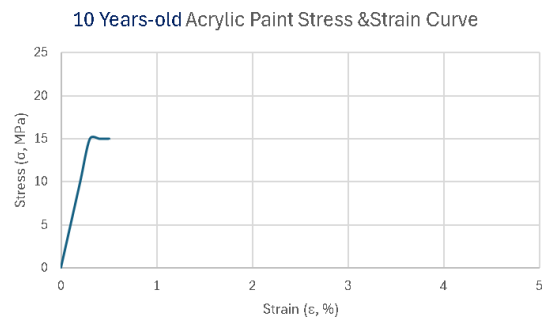


Fig. 22. Hypothetical ten-years-old acrylic paint stress-strain curve

When it comes to brittle material, it tends to withstand compressive stress better than tensile stress. As seen in Figs. 7 and 8 above, there are many more cracks in Fig. 7 where tensile stress is applied. The difference in crack occurrence rates between the outer and inner surfaces of the curvature can also be seen as an effect of the current brittle paint layer's ability to withstand compressive stress better than tensile stress.¹⁰

4.3 Flaws and Crack Expansion (Griffith's Theory)

According to Griffith's Theory, when defects such as microscopic cracks or flaws exist in brittle materials, they lead to stress concentration.¹¹ When external stress is applied, it amplifies significantly at these crack tips, which have a small radius, resulting in crack expansion. In the case of this object, defects from the manufacturing process may be present, or microcracks initiated by various stresses appear to have expanded (Fig.9). Furthermore, as described above, the paint on this object has become brittle, making cracks more likely to propagate.

5. Conservation Approaches

5.1 Interventive and Preventative Conservation

The conservation process of a damaged object generally involves four stages: condition analysis,

¹⁰ William D. Allister and David G. Rethwisch, *Materials Science and Engineering*, 173.

¹¹ Chin-Teh Sun, "Griffith Theory of Fracture," in *Fracture Mechanics* (United States: Elsevier Science & Technology, 2011), 14.

cleaning, reintegration, and preservation. Considering the most relevant phase to the material failure, selecting the appropriate medium for reintegration is critical. For the chandelier, lacking precise information about the metal and paint properties limits specific conservation recommendations. However, successful conservation still depends on the paint layer's material properties.

Selection of Paint

When selecting paint, three key factors must be considered: flexibility, adhesion power, and thermal expansion characteristics. Cracks and peeling likely result from insufficient paint flexibility relative to metal movement. The conservation material requires balanced flexibility. If too brittle, it may crack under minor movements, while if too flexible, it may lose shape during drying. Strong adhesion power is essential to prevent detachment under environmental stresses like humidity, temperature fluctuations, or vibrations. Additionally, choosing paint with a coefficient of thermal expansion (CTE) close to that of the metal minimizes fatigue from differential expansion. Considering the general condition, **Epoxy resin** paint could be suitable for the metal object.¹²

Adding a Coating Layer

Another effective approach is applying a coating over the paint layer to prevent cracks from static forces like residual stresses or dynamic forces such as impacts. Experimental evidence (Fig.23) shows that coating a painted metal object can delay corrosion onset.^{13 14}

The same considerations for paint including flexibility, adhesion power, and thermal expansion, apply to coating materials, with the additional requirement of compatibility with the paint layer. As a coating layer, **Paraloid B-72** provides effective metal conservation through corrosion protection, flexibility, good adhesion, and stability.¹⁵ However, as same as the paint selection, the coating material's CTE still should be compared with the CTE of the metal and paint to ensure compatibility.

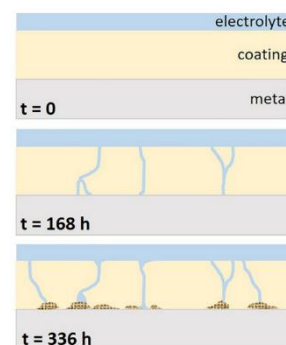


Fig. 23. Schematic representation of the evolution of a coated metal upon aging.

Preservation

After conservation, proper storage requires controlling temperature (18-22°C) and relative humidity (40-55%), avoiding direct sunlight, and using LED bulbs to minimize heat generation. Preventive measures should focus on minimizing impacts that could cause deformation. When handling or installing the object, wearing gloves provides shock absorption, and wrapping each part in cloth ensures shock-proof storage.

6. Conclusion and Further Research

¹² SinoGraceChem, "Cohesion and Thermal Expansion."

¹³ María Teresa Molina, Emilio Cano, and Blanca Ramírez-Barat, "Testing Protective Coatings for Metal Conservation: The Influence of the Application Method," *Heritage Science* 11, no. 1 (2023):8.

¹⁴ In the test setup shown in the accompanying diagram, the effectiveness of the coating is measured based on how long it would take until an electrolyte containing ions reaches the metal through the cracks. While the time it takes for the metal to come into contact with a corrosive environment after coating application may vary depending on environmental conditions and material properties, this experiment effectively demonstrates the coating's typical rust-prevention benefits.

¹⁵ María Teresa Molina, Emilio Cano, and Blanca Ramírez-Barat, "Testing Protective Coatings for Metal Conservation: The Influence of the Application Method," *Heritage Science* 11, no. 1 (2023): 4,5,7.



Understanding mechanical property failure mechanisms is crucial for designing safer and more durable conservation strategies. This report analyzed various mechanical failures, exploring their causes, effects, and potential mitigation approaches. The main failure modes discussed are caused by residual stresses within the metal, the coefficient of thermal expansion, corrosion, stress concentrations, aging of paint, and flaws that cause cracks to propagate. Based on these findings, effective conservation strategies should focus on proper material selection, the application of protective coating layers, and ensuring that the object is stored in a controlled environment to minimize further deterioration.

In future research, chemical analysis of material properties and adhesion strength analysis would be beneficial, particularly in studies examining the interactions between paint and metal substrates. These investigations could provide crucial insights into failure mechanisms at material interfaces in this object, ultimately reducing risks and enhancing structural integrity.

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